Multi-layer weighted social network model

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Abstract

Recent empirical studies using large-scale datasets have validated the Granovetter hypothesis on the structure of the society in that there are strongly wired communities connected by weak ties. However, as interaction between individuals takes place in diverse contexts, these communities turn out to be overlapping. This implies that the society has a multi-layered structure, where the layers represent the different contexts. We present a multi-layer weighted social network model, which yields enhanced overlapping communities and, at the same time, the Granovetterian weight-topology relation.

Keyword: social network, Granovetter structure, overlapping communities

The abundance of data due to the rapid development of the information and communication technology (ICT) has generated entirely new, multidisciplinary approaches in social sciences. Among a broad range of ICT data, mobile phone data have a special role in this endeavor as much of the inter-personal communication runs today over mobile phones. In this mobile call network the famous Granovetter hypothesis about the "strength of weak ties" turned out to be correct: The society consists of communities, which are strongly wired and these communities are then connected by weak ties, thus playing an important role to hold society together.

To understanding the mechanisms leading to this Granovetterian structure, a model which incorporates basic link-formation process between individuals was proposed [1]. With simple rules called local and global attachments, the complex Granovetterian weight-topology relation of social networks could be successfully reproduced.

Another important aspect of the socieal network is that a person can be in different types of relationships, like kinship, collaboration, friendship, etc. Moreover, people are switching their social contexts and communication channels depending on the occasions, and the social network should strongly dependent on the context. To handle these aspects, it is necessary to represent the social networks as a multi-layer or multiplex network, where each layer corresponds to a different type of relationship. The aim of our study is to investigate the possibilities to model the combination of the multi-layer structure of the society with the Granovetterian relationship between tie strengths and topology.

To model this structure, we begin with a single-layer weighted social network (WSN) model showing the Granovetterian structure [1]. We find that when merging such WSN models, sufficient amount of inter-layer correlation is needed to maintain the relationship between topology and link weights but these correlations destroy the enhancement in the community overlap due to multiple layers. To resolve this, we devise a geographic multi-layer WSN model, where the indirect inter-layer correlations due to the geographic constraints of individuals enhance the overlaps between the communities and, at the same time, the Granovetterian structure is preserved [2].

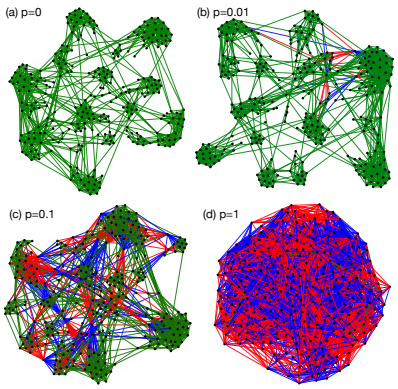


Figure1: Snapshots of the correlated multi-layer WSN. Granovetterian structure is destroyed as inter-layer correlation gets weaker (from (a) to (d)).

References

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